

For task 4, https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-05.csv dataset is used

Task 4. Write MapReduce codes to perform the tasks using the files you've downloaded on your EMR Instance:

a. Which vendors have the most trips, and what is the total revenue generated by that vendor?

```
[hadoop@ip-172-31-85-87 ~]$ python mrtask_a.py yellow_tripdata_2017-05.csv > mrtask_a_result.txt
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_a.hadoop.20250106.222247.266089
Running step 1 of 2...
Running step 2 of 2...
job output is in /tmp/mrtask_a.hadoop.20250106.222247.266089/output
Streaming final output from /tmp/mrtask_a.hadoop.20250106.222247.266089/output...
Removing temp directory /tmp/mrtask_a.hadoop.20250106.222247.266089...
[hadoop@ip-172-31-85-87 ~]$ cat mrtask_a_result.txt
"2"      92896777.54522054
[hadoop@ip-172-31-85-87 ~]$
```

Vendor 2 i.e. VeriFone Inc. has most trips and revenue generated is 92896777.545

b. Which pickup location generates the most revenue?

```
[hadoop@ip-172-31-85-87 ~]$ python mrtask_b.py yellow_tripdata_2017-05.csv > mrtask_b_result.txt
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_b.hadoop.20250106.223133.004653
Running step 1 of 2...
Running step 2 of 2...
job output is in /tmp/mrtask_b.hadoop.20250106.223133.004653/output
Streaming final output from /tmp/mrtask_b.hadoop.20250106.223133.004653/output...
Removing temp directory /tmp/mrtask_b.hadoop.20250106.223133.004653...
[hadoop@ip-172-31-85-87 ~]$ cat mrtask_b_result.txt
"132"    14040591.220016211
[hadoop@ip-172-31-85-87 ~]$
```

Pickup location on 132 generate most revenue (14040591.22)

c. What are the different payment types used by customers and their count? The final results should be in a sorted format.

```
[hadoop@ip-172-31-85-87 ~]$ python mrtask_c.py yellow_tripdata_2017-05.csv > mrtask_c_result.txt
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_c.hadoop.20250106.223635.007449
Running step 1 of 3...
Running step 2 of 3...
Running step 3 of 3...
job output is in /tmp/mrtask_c.hadoop.20250106.223635.007449/output
Streaming final output from /tmp/mrtask_c.hadoop.20250106.223635.007449/output...
Removing temp directory /tmp/mrtask_c.hadoop.20250106.223635.007449...
[hadoop@ip-172-31-85-87 ~]$ cat mrtask_c_result.txt
"1"      6780947
"2"      3250362
"3"      55027
"4"      15791
[hadoop@ip-172-31-85-87 ~]$
```

Different payment types used by customers with their count

1= Credit card 6780947 times

2= Cash 3250362 times

3= No charge 55027 times

4= Dispute 15791 times

Payment through credit card has been done more often.

d. What is the average trip time for different pickup locations?

```
[hadoop@ip-172-31-85-231 ~]$ python mrtask_d.py yellow_tripdata_2017-05.csv mrtask_d
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_d.hadoop.20250107.000805.402626
Running step 1 of 1...
job output is in /tmp/mrtask_d.hadoop.20250107.000805.402626/output
Streaming final output from /tmp/mrtask_d.hadoop.20250107.000805.402626/output...
"1"      12.681244881244885
"10"     56.4414970815586
"100"    16.301056898951547
"101"    10.8640350877193
"102"    15.480072463768114
"105"    22.727777777777778
"106"    14.046902507889056
"107"    14.914703800523975
"108"    17.747747747747745
"109"    13.047222222222222
"11"     11.318787878787878
"111"    15.347685185185185
"112"    15.342635286583413
"113"    15.655663780443364
"114"    16.265089572862763
"115"    9.849019607843136
"116"    15.213805066967888
"117"    9.020000000000001
"118"    16.645833333333336
"119"    14.638700564971753
"12"     25.11584025215252
"120"    12.665972222222223
"121"    16.845833333333333
"122"    11.954761904761904
"123"    12.698253968253967
"124"    35.719855072463766
"125"    16.582152788143137
"126"    32.175581395348836
"127"    17.629476052716164
"128"    15.57929292929293
"129"    14.331578947368426
"13"     20.914534001214356
"130"    37.98320691108302
"131"    11.690000000000003
"132"    46.805229740819875
"133"    17.532022849462365
"134"    16.221703296703296
"135"    15.368173258003768
"136"    9.693303571428572
"137"    14.412591446828932
"138"    41.79266819772832
"139"    18.82820512820513
"14"     18.910311918063318
"140"    14.544783434852864
"141"    12.887731943471524
"142"    14.724678434186814
"143"    13.87881130774534
```

"144"	17.422410667367306	"200"	11.855392156862747
"145"	13.919851029794044	"201"	0.49374999999999997
"146"	15.779948824343021	"202"	15.435855263157896
"147"	13.003051643192489	"203"	19.28479532163743
"148"	16.994242069527864	"204"	0.47999999999999997
"149"	13.134482758620688	"205"	18.713008130081302
"15"	14.868589743589745	"206"	11.648809523809524
"150"	22.067647058823532	"207"	8.778993536472758
"151"	13.72286961525267	"208"	15.355907172995781
"152"	15.034273318872025	"209"	20.52999086340794
"153"	12.776791277258567	"21"	16.197798742138364
"154"	22.930303030303033	"210"	18.476602564102567
"155"	23.847222222222218	"211"	17.805909853173578
"156"	17.777777777777778	"212"	24.377777777777773
"157"	24.96510297482838	"213"	20.40786163522013
"158"	17.056993321982915	"214"	8.220833333333333
"159"	12.977884615384616	"215"	56.68466165413534
"16"	11.23930817610063	"216"	34.08083333333333
"160"	17.381399631675876	"217"	18.35507590132827
"161"	16.501390496486135	"218"	17.684177215189873
"162"	15.5755374379993	"219"	53.02001915708813
"163"	17.007285407402943	"22"	22.21085626911315
"164"	16.135913909640667	"220"	11.665315315315317
"165"	23.146643518518516	"221"	19.811904761904763
"166"	16.128181641420127	"222"	19.394444444444442
"167"	11.731709401709402	"223"	17.158481387478847
"168"	11.912370135587249	"224"	14.423218777619141
"169"	12.008282828282828	"225"	15.393991912189485
"17"	14.360986719350239	"226"	17.062662041086035
"170"	15.23280819184737	"227"	15.537459546925568
"171"	11.867179487179486	"228"	18.118900966183574
"172"	29.29583333333333	"229"	13.729634370452187
"173"	12.247398589065257	"23"	3.605128205128205
"174"	12.091085271317832	"230"	18.16775177926668
"175"	11.432758620689656	"231"	17.842905702290793
"176"	0.6	"232"	17.80712673224909
"177"	20.955774278215223	"233"	16.130618672379082
"178"	8.291562500000001	"234"	15.740532219114375
"179"	15.1499705882353	"235"	21.51443736730361
"18"	12.653225806451612	"236"	13.552552898225786
"180"	27.183506944444446	"237"	12.943728784429947
"181"	15.849603785417022	"238"	13.635731925509624
"182"	12.2456	"239"	13.766637412862702
"183"	11.927777777777779	"24"	14.086371798958675
"184"	7.346666666666666	"240"	16.91818181818182
"185"	15.095833333333333	"241"	15.555228758169934
"186"	17.3607280512866	"242"	13.335582822085891
"187"	1.425	"243"	17.764382566585954
"188"	15.907792044310169	"244"	17.97972746905131
"189"	16.631431947577614	"245"	23.330555555555552
"19"	10.768354430379746	"246"	16.162821706410472
"190"	26.461985335589397	"247"	16.876773493573694
"191"	13.486574074074074	"248"	13.558333333333332
"192"	13.586842105263157	"249"	15.335284107159861
"193"	12.075183464619935	"25"	17.17143186640833
"194"	27.354013503375846	"250"	14.828939393939397
"195"	22.571194503171245	"251"	2.1999999999999997
"196"	18.181564465408805	"252"	22.431190476190476
"197"	18.036302681992343	"253"	25.680555555555554
"198"	13.806195462478186	"254"	11.153439153439153
"2"	42.2	"255"	18.58691486586808
"20"	17.141091954022986	"256"	15.882166210462294

"257"	16.62408789386401
"258"	26.296310432569975
"259"	10.456666666666669
"26"	11.717295597484279
"260"	17.798491342362418
"261"	22.742345329876898
"262"	13.453498021610727
"263"	12.558972943698416
"264"	15.660701878308133
"265"	10.097814982973892
"27"	9.104166666666668
"28"	33.768
"29"	57.221774193548384
"3"	17.91923076923077
"30"	14.595833333333333
"31"	22.172916666666667
"32"	45.23925925925926
"33"	19.48258199014998
"34"	15.771012544802865
"35"	12.903932584269663
"36"	16.217423442449842
"37"	15.67785183038733
"38"	105.58416666666668
"39"	17.3040625
"4"	15.943539005698556
"40"	15.923511762519249
"41"	13.047298730655545
"42"	13.060410811865738
"43"	16.67567976383065
"44"	9.504166666666666
"45"	19.516391550098202
"46"	29.257407407407406
"47"	15.23495145631068
"48"	15.306237295695665
"49"	14.44308345010119
"5"	0.3
"50"	15.42537099704824
"51"	11.96068376068376
"52"	18.217088839941262
"53"	20.58723958333333
"54"	14.719335511982568
"55"	47.3187265917603
"56"	23.806329113924047
"57"	15.616666666666667
"58"	5.333333333333333
"59"	19.616666666666664
"6"	7.790277777777777
"60"	12.961559139784944
"61"	15.251830481496228
"62"	13.879747675962818
"63"	65.07303921568626
"64"	11.29107142857143
"65"	18.156792744967923
"66"	18.57747506019952
"67"	13.053431372549017
"68"	15.859693387283983
"69"	19.92417624521073
"7"	13.984743997175142
"70"	26.37702202480104
"71"	12.455172413793102
"72"	17.463793103448275
"73"	22.723684210526315

```
"74"    12.525644805036702
"75"    13.519075992588938
"76"    18.701871980676327
"77"    16.11434108527132
"78"    11.448753894080996
"79"    15.650715220659471
"8"     22.969314641744553
"80"    16.722509667955716
"81"    10.016666666666667
"82"    15.163276362823948
"83"    16.827437223042836
"84"    5.575
"85"    15.723731138545956
"86"    14.700000000000001
"87"    22.17575468660486
"88"    23.09566049232271
"89"    17.87714633298575
"9"     153.69927536231884
"90"    14.588625022901905
"91"    23.28269720101781
"92"    14.84195171026157
"93"    38.06967213114755
"94"    11.647380952380951
"95"    18.848467302452317
"96"    18.588888888888885
"97"    16.081618563620673
"98"    13.002500000000001
Removing temp directory /tmp/mrtask_d.hadoop.20250107.000805.402626...
```

e. Calculate the average tips to revenue ratio of the drivers for different pickup locations in sorted format.

```
[hadoop@ip-172-31-85-231 ~]$ python mrtask_e.py yellow_tripdata_2017-05.csv > mrtask_e_result.txt
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_e.hadoop.20250106.234322.381308
Running step 1 of 1...
job output is in /tmp/mrtask_e.hadoop.20250106.234322.381308/output
Streaming final output from /tmp/mrtask_e.hadoop.20250106.234322.381308/output...
Removing temp directory /tmp/mrtask_e.hadoop.20250106.234322.381308...
[hadoop@ip-172-31-85-231 ~]$ cat mrtask_e_result.txt
"1"      0.11961918264336044
"10"     0.1031314953272488
"100"    0.10054811858030266
"101"    0.1538484771604267
"102"    0.09686867923894887
"105"    0.07175925925925926
"106"    0.11203391656570175
"107"    0.11929564967810695
"108"    0.12120884627098828
"109"    0.23737785016286642
"11"     0.058159884648828326
"111"    0.077659754354142
"112"    0.10900187462682069
"113"    0.11769752457713911
"114"    0.11571668143567702
"115"    0.1014877461571702
"116"    0.09140144834181262
"117"    0.029865191166550988
"118"    0.03324247058041353
"119"    0.06439098561727422
"12"     0.09033344274559964
"120"    0.14485426649668087
"121"    0.06025386778698323
"122"    0.035274937009041045
"123"    0.0750228876958997
"124"    0.09886020334918795
```


"125"	0.1223467893609015	"181"	0.11383678943060177
"126"	0.05493403420192623	"182"	0.04935382990636337
"127"	0.08833111068775	"183"	0.04058310746465461
"128"	0.10511320192841678	"184"	0.03374233128834356
"129"	0.06647065796213625	"185"	0.07860584480628201
"13"	0.12390141102889671	"186"	0.10968385132719934
"130"	0.1073558396219219	"187"	0.03211303789338471
"131"	0.10384807204137783	"188"	0.07172143661981017
"132"	0.1025660513460536	"189"	0.10932535246060386
"133"	0.09147281347725217	"19"	0.11032972645756875
"134"	0.10314438578934823	"190"	0.10117750519544694
"135"	0.0785122214655781	"191"	0.05438239409744308
"136"	0.032600578121018825	"192"	0.0720589495330306
"137"	0.11253077758026406	"193"	0.03922346473669742
"138"	0.1304834446631976	"194"	0.12076731965739748
"139"	0.025278845473624637	"195"	0.09492841545380133
"14"	0.09666329734829485	"196"	0.09427848086047703
"140"	0.1110704470853594	"197"	0.10215364731577099
"141"	0.11186974122790275	"198"	0.10078643815066789
"142"	0.1131558975317282	"2"	0.11989767545323099
"143"	0.11473162975887714	"20"	0.0736013986013986
"144"	0.1105272721674073	"200"	0.13187606166264348
"145"	0.09866671202718473	"201"	0.132044713553796
"146"	0.09229805528093973	"202"	0.09899140045991586
"147"	0.04671430427439263	"203"	0.048087416271543615
"148"	0.11347366987186253	"204"	0.0684931506849315
"149"	0.12259087582337157	"205"	0.07523246599344224
"15"	0.12542820971105154	"206"	0.08833170762908359
"150"	0.10633375877472878	"207"	0.0815078125388226
"151"	0.10794854174655912	"208"	0.08291166272142231
"152"	0.08497213797829244	"209"	0.11297948250644907
"153"	0.08409909578769388	"21"	0.06386360722984234
"154"	0.07400473993394348	"210"	0.07934380776340108
"155"	0.09130948944530835	"211"	0.11310320504679515
"156"	0.07496251874062969	"212"	0.06409624702307627
"157"	0.11154334990834797	"213"	0.11780938978546256
"158"	0.1177428105347338	"214"	0.02993836219547989
"159"	0.04880437380957894	"215"	0.10353565292295142
"16"	0.1296825635625094	"216"	0.11008942973165864
"160"	0.09369565599255213	"217"	0.07868790199337758
"161"	0.09989106877197958	"218"	0.19235819846724786
"162"	0.11962265897953572	"219"	0.1049626881741011
"163"	0.11101476605573193	"22"	0.05041866234795492
"164"	0.11138449806107008	"220"	0.056836272960188075
"165"	0.07196634804086764	"221"	0.10844779603195978
"166"	0.11344656333427505	"222"	0.11278732656864776
"167"	0.0356042774025466	"223"	0.09822645175828154
"168"	0.050410277807923884	"224"	0.11425442486544808
"169"	0.11765956298165817	"225"	0.08099702729092222
"17"	0.08304806476027186	"226"	0.08354577861128779
"170"	0.11737627919224174	"227"	0.08721017303002598
"171"	0.04416689447785676	"228"	0.09978962780363482
"172"	0.27196083763937995	"229"	0.11203502738122283
"173"	0.0662888212150767	"23"	0.08146106133700898
"174"	0.09292588232864547	"230"	0.10662429938743652
"175"	0.08816570904350345	"231"	0.11828076056384894
"176"	0.19982277359326542	"232"	0.10577578661446632
"177"	0.09851238280057963	"233"	0.11514074932415078
"178"	0.03941897087008663	"234"	0.12096822968790891
"179"	0.08647112167836499	"235"	0.0557415553911836
"18"	0.07413049841666887	"236"	0.11249379678942588
"180"	0.10306335321889983	"237"	0.1096331664258742

"238"	0.10834063914173245	"43"	0.10407835112043434
"239"	0.11534544071014695	"44"	0.07601977750309023
"24"	0.10648126258555479	"45"	0.0983846147461719
"240"	0.10447949050358085	"46"	0.12534064929126656
"241"	0.06993302518630319	"47"	0.03340712045748017
"242"	0.07890211996979177	"48"	0.10582758808855866
"243"	0.09530565669975881	"49"	0.09681013151351817
"244"	0.09546944114505117	"5"	0.711159737417943
"245"	0.0	"50"	0.10923919835836489
"246"	0.11995315301676546	"51"	0.054687879641958155
"247"	0.07335984971767694	"52"	0.13055804576734864
"248"	0.052623959464350334	"53"	0.09939808622286242
"249"	0.11923576897943587	"54"	0.11968717470048443
"25"	0.11052464311894397	"55"	0.10775881095465412
"250"	0.043282871250452847	"56"	0.09187656958656626
"251"	0.0019544811624019544	"57"	0.0982393882217696
"252"	0.08558204423759197	"58"	0.023456558453743663
"253"	0.06711331976762965	"59"	0.022958827634333566
"254"	0.0972480119905075	"6"	0.04668850781897014
"255"	0.11601004961517414	"60"	0.028830566964685292
"256"	0.1093043993422745	"61"	0.09614549547751258
"257"	0.10482648999719887	"62"	0.07257887627997188
"258"	0.15786461137400468	"63"	0.07430008293013912
"259"	0.12318663572582067	"64"	0.07777953179990528
"26"	0.05332139249601685	"65"	0.1064885388809956
"260"	0.07205229212930792	"66"	0.11474873960246652
"261"	0.10801526749738075	"67"	0.08016136157322679
"262"	0.11380613560114887	"68"	0.1143179862016906
"263"	0.1142608495023124	"69"	0.03888023995673477
"264"	0.11230316290901997	"7"	0.08055330999297121
"265"	0.10968911731785466	"70"	0.10558719618955761
"27"	0.07044812927929389	"71"	0.057126398156593525
"28"	0.10315928607679181	"72"	0.12979482386561036
"29"	0.04496955778040933	"73"	0.06660273315751619
"3"	0.0352269585440825	"74"	0.07309332094094213
"30"	0.431189363995688	"75"	0.09233854474158283
"31"	0.09241457441314165	"76"	0.15699972823625333
"32"	0.07388668844623049	"77"	0.03758990781571329
"33"	0.1204236725267228	"78"	0.03340662069771519
"34"	0.11469105842822479	"79"	0.11726023735794265
"35"	0.17581828771112398	"8"	0.1128890978373831
"36"	0.10412970427360184	"80"	0.11230199593086634
"37"	0.103438070866592	"81"	0.1314140628199353
"38"	0.03832478224555542	"82"	0.0640252588354194
"39"	0.11607313499394789	"83"	0.060647576771211555
"4"	0.10621226787883002	"84"	0.0006641576267434137
"40"	0.11953106379554157	"85"	0.0568739741017691
"41"	0.09274584418153507	"86"	0.03836325514349365
"42"	0.07096965920546744	"87"	0.12371478220911764
"43"	0.10407835112043434	"88"	0.11535120965220538
"44"	0.07601977750309023	"89"	0.08902658805206363
"45"	0.0983846147461719	"9"	0.07875511622519114
"46"	0.12534064929126656	"90"	0.11768850776024835
"47"	0.03340712045748017	"91"	0.10308391768201007
"48"	0.10582758808855866	"92"	0.056615381413745704
"49"	0.09681013151351817	"93"	0.1071444344921823
"5"	0.711159737417943	"94"	0.038084346952212626
"50"	0.10923919835836489	"95"	0.07928530356245143
"51"	0.054687879641958155	"96"	0.09975598325818912
"52"	0.13055804576734864	"97"	0.09868461512594295
"53"	0.09939808622286242	"98"	0.06067388225420083
"54"	0.11968717470048443		[hadoop@ip-172-31-85-231 ~]\$

f. How does revenue vary over time? Calculate the average trip revenue per month - analysing it by hour of the day (day vs night) and the day of the week (weekday vs weekend).

```
[hadoop@ip-172-31-85-231 ~]$ python mrtask_f.py yellow_tripdata_2017-05.csv > mrtask_f_result.txt
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_f.hadoop.20250106.235218.122386
Running step 1 of 1...
job output is in /tmp/mrtask_f.hadoop.20250106.235218.122386/output
Streaming final output from /tmp/mrtask_f.hadoop.20250106.235218.122386/output...
Removing temp directory /tmp/mrtask_f.hadoop.20250106.235218.122386...
[hadoop@ip-172-31-85-231 ~]$ cat mrtask_f_result.txt
[5, 0, 0]      19.09149097120672
[5, 0, 1]      19.42010595117413
[5, 0, 2]      17.686158986637274
[5, 0, 3]      17.25599944664103
[5, 0, 4]      18.099999027301898
[5, 0, 5]      17.105128026116333
[5, 0, 6]      15.658808411947584
[5, 1, 0]      17.87588900705367
[5, 1, 1]      19.429909785928686
[5, 1, 2]      17.373021160869822
[5, 1, 3]      16.436848288253056
[5, 1, 4]      18.217479700823546
[5, 1, 5]      16.275959370214274
[5, 1, 6]      15.141293067405417
[5, 10, 0]     16.241741228653577
[5, 10, 1]     16.639627234011154
[5, 10, 2]     27.168365753878387
[5, 10, 3]     16.499299606770162
[5, 10, 4]     16.48680345506514
[5, 10, 5]     14.029692293163937
[5, 10, 6]     14.908801744148995
[5, 11, 0]     16.680433054406084
[5, 11, 1]     17.19449236461242
[5, 11, 2]     16.92334503331534
[5, 11, 3]     18.596232180453956
[5, 11, 4]     16.728098769455887
[5, 11, 5]     14.021549507826318
[5, 11, 6]     14.98238805727666
[5, 12, 0]     16.480360929345835
[5, 12, 1]     17.22963705731718
[5, 12, 2]     17.05563095494162
[5, 12, 3]     17.975730053716557
[5, 12, 4]     17.69103689287293
[5, 12, 5]     14.499792055020734
[5, 12, 6]     15.27335388620518
[5, 13, 0]     16.795085865627826
[5, 13, 1]     17.27948235089487
[5, 13, 2]     17.774827762048346
[5, 13, 3]     18.595673825475238
[5, 13, 4]     18.205408289793773
[5, 13, 5]     14.99034880898968
[5, 13, 6]     15.937601258660894
[5, 14, 0]     17.20858440343585
[5, 14, 1]     17.34785479975938
[5, 14, 2]     18.110961353384074
```

[5, 14, 3]	19.033494081619924	[5, 22, 2]	16.730750002516206
[5, 14, 4]	18.613387909146194	[5, 22, 3]	17.699894119517573
[5, 14, 5]	15.336584485461453	[5, 22, 4]	16.72930392121595
[5, 14, 6]	17.128896489169726	[5, 22, 5]	15.161717603507384
[5, 15, 0]	16.91500182066798	[5, 22, 6]	17.47747548027094
[5, 15, 1]	17.307682451791745	[5, 23, 0]	18.734166782523527
[5, 15, 2]	17.75765307206797	[5, 23, 1]	17.364087663335088
[5, 15, 3]	18.750789042356082	[5, 23, 2]	17.252883134436264
[5, 15, 4]	18.479680845892585	[5, 23, 3]	18.14697136778204
[5, 15, 5]	15.509464318824842	[5, 23, 4]	17.248078089078962
[5, 15, 6]	17.056592424577616	[5, 23, 5]	15.846046419400455
[5, 16, 0]	18.26141013123625	[5, 23, 6]	18.86490867580518
[5, 16, 1]	18.859940648116382	[5, 3, 0]	17.010436833469626
[5, 16, 2]	19.6542708916958	[5, 3, 1]	17.389016241913282
[5, 16, 3]	20.04885999577759	[5, 3, 2]	16.62262594124502
[5, 16, 4]	19.834795454141705	[5, 3, 3]	16.37661024166937
[5, 16, 5]	15.635747754243198	[5, 3, 4]	17.679739001270963
[5, 16, 6]	17.32405836477559	[5, 3, 5]	16.147548793518364
[5, 17, 0]	17.555959492507064	[5, 3, 6]	15.822056911276535
[5, 17, 1]	17.747241962602008	[5, 4, 0]	21.292999725297417
[5, 17, 2]	18.3960718339617	[5, 4, 1]	20.068907006970854
[5, 17, 3]	18.887732509348027	[5, 4, 2]	19.732507288628188
[5, 17, 4]	18.411963137349552	[5, 4, 3]	20.829968819597944
[5, 17, 5]	15.359309303734426	[5, 4, 4]	20.915669272681466
[5, 17, 6]	17.4439645341407	[5, 4, 5]	17.825446810289648
[5, 18, 0]	16.500203407324047	[5, 4, 6]	17.38188976038157
[5, 18, 1]	16.477160631363912	[5, 5, 0]	21.593018355659638
[5, 18, 2]	17.053779035240108	[5, 5, 1]	18.81002788103778
[5, 18, 3]	17.221604263570683	[5, 5, 2]	18.894916666663246
[5, 18, 4]	16.83762356868249	[5, 5, 3]	19.845702192891927
[5, 18, 5]	14.73267798036185	[5, 5, 4]	21.227005213121405
[5, 18, 6]	16.363879207901295	[5, 5, 5]	20.992848200309997
[5, 19, 0]	16.117071206744605	[5, 5, 6]	21.125313003037945
[5, 19, 1]	15.933338279383616	[5, 6, 0]	16.36666989228269
[5, 19, 2]	16.270620125859278	[5, 6, 1]	14.978345859721552
[5, 19, 3]	16.472398718229226	[5, 6, 2]	14.767823020740611
[5, 19, 4]	16.161423076937226	[5, 6, 3]	15.288311117881104
[5, 19, 5]	14.486719040663449	[5, 6, 4]	16.389897123112565
[5, 19, 6]	16.246780024174765	[5, 6, 5]	19.32208822201787
[5, 2, 0]	17.069302560401354	[5, 6, 6]	19.992739240104342
[5, 2, 1]	17.102999568033265	[5, 7, 0]	15.307352827781946
[5, 2, 2]	16.490724900576904	[5, 7, 1]	14.68863340021126
[5, 2, 3]	15.885527411273001	[5, 7, 2]	14.46050088621336
[5, 2, 4]	17.510186185909916	[5, 7, 3]	14.619068724817286
[5, 2, 5]	15.787822292178655	[5, 7, 4]	14.939835919422332
[5, 2, 6]	15.219828879150024	[5, 7, 5]	16.300197449100637
[5, 20, 0]	16.278329889820224	[5, 7, 6]	18.066767322725
[5, 20, 1]	15.92229313006401	[5, 8, 0]	15.534355540424196
[5, 20, 2]	15.999283947134947	[5, 8, 1]	15.310034719953286
[5, 20, 3]	16.363236084025758	[5, 8, 2]	14.991955112543714
[5, 20, 4]	16.041364614564298	[5, 8, 3]	15.095595443793671
[5, 20, 5]	15.008297507476271	[5, 8, 4]	15.144876788509523
[5, 20, 6]	16.514845325711754	[5, 8, 5]	14.67262836614724
[5, 21, 0]	16.81774921469818	[5, 8, 6]	15.685100979728976
[5, 21, 1]	16.388015828166143	[5, 9, 0]	15.79006959000546
[5, 21, 2]	16.231406528150977	[5, 9, 1]	15.809253019981295
[5, 21, 3]	17.121041382564638	[5, 9, 2]	15.622612773623201
[5, 21, 4]	16.579383172574527	[5, 9, 3]	15.764076749004406
[5, 21, 5]	15.030104427307885	[5, 9, 4]	15.912008246957104
[5, 21, 6]	16.83408532267027	[5, 9, 5]	13.814643752241684
[5, 22, 0]	17.337623268018195	[5, 9, 6]	14.779713618189996
[5, 22, 1]	17.00356693816382	[hadoop@ip-172-31-85-231 ~]\$	